Book-Crossing Community Insight: Designing an Efficient Data Engineering Pipeline for Reading Community Analytics

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Abstract—The Book-Crossing dataset suffers from data problems such as three CSV files that are not consistent with 140,990 records. There are many data quality issues like missing primary keys, duplicates, and wrong values for demographics. There are no rules that make sure that the different files are connected correctly. We describe the end-to-end design and implementation of an automated Extract, Transform, Load (ETL) data pipeline that solves these challenges. The pipeline uses Pentaho Data Integration (PDI) to load raw data into a MySQL raw layer, perform a three-level quality validation framework (structural, domain and referential integrity checks) on a staging layer and load verified data into a star schema data warehouse containing one fact table (fact_ratings) and two dimension tables (dim_book, dim_user). A single Pentaho Job kicks off the whole pipeline, triggering the nine transformational steps in order of dependency and has been built to be completely idempotent. This data warehouse has been populated with 111585 validated rating transactions for 496 unique books and 28909 unique users to build an interactive dashboard in Power BI, providing insights into the popularity of books, engagement of authors, demographics of readers and global geographic distribution in the community." Tool evaluation indicates the main scalability limiting factors are the full load ingestion strategy and MySQL's row-oriented engine, and the main future directions are incremental loading and cloud-native columnar storage.

Keywords— data pipeline, ETL, Pentaho Data Integration, data warehouse, star schema, data quality, MySQL, Power BI, Book Crossing, community analytics

I. INTRODUCTION

Online literary sites have become virtual communities where users can discover, review, and rate books, sharing their reading experience. These activities produce large amounts of user interaction data. If processed correctly, this data can reveal the patterns of reading behavior and help to determine what books are the most popular among readers.

One of the most used datasets in this type of analysis is the Book-Crossing dataset. It contains user profiles, book metadata and explicit rating records collected from readers from a number of countries. Its size and richness have made it a popular reference for studies of online reading communities.

But the dataset is not easily available for analysis. It is distributed in three separate CSV files for users, books, and ratings. There is no guaranteed relational integrity between the three files, It also has several data quality problems, such as missing ISBN values, duplicate records, unrealistic age values, and inconsistent publication information. These problems must be solved first, otherwise any analysis based on this data is likely to lead to misleading conclusions rather than useful insights.

Most of the existing studies on the Book-Crossing dataset have focused on recommendation systems or user behavior analysis with little attention to how the dataset should be cleaned and prepared a priori through a structured Extract, Transform, Load (ETL) process. This paper takes this gap as the starting point, designs, implements an end-to-end automated ETL pipeline for the Book-Crossing dataset. The proposed pipeline employs Pentaho Data Integration (PDI) and MySQL in a three-tier architecture: a raw layer to keep the original data, a staging layer for data cleansing and transformation, and a data warehouse structured in a star schema. The data is visualized through an interactive Power BI dashboard that allows for the analysis of book popularity, author engagement, reader demographics, and geographic distribution.

This research has several key contributions: (1) the development of a three-tier ETL architecture that enhances data quality before analysis; (2) the creation of automated Pentaho Jobs that allow for repeatable and idempotent ETL workflows; (3) the building of a star schema data warehouse using surrogate keys to facilitate efficient analytical queries; and (4) the development of an interactive Power BI dashboard for multidimensional analysis of the Book-Crossing community. The rest of this paper is organized as follows. Section II reviews relevant literature. Section III outlines system design and research methods. Section IV details the implementation of the proposed workflow. The experimental results are given in Section V and the paper is concluded in Section VI with a discussion of future work.

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II. LITERATURE REVIEW

A. Data Warehouses and Star Schema Architecture

The development of a data warehouse is intended to integrate data from various fragmented transactional sources into a single centralized repository to support multidimensional analysis needs. Unlike Online Transaction Processing (OLTP) databases, which are optimized for daily operations involving dense data, data warehouse architecture is designed to minimize computational load when handling complex reporting queries.

As one of the primary dimensional modeling approaches, Kimball and Ross introduced the star schema methodology [2]. This architectural design clearly separates the data storage structure into a single central fact table surrounded by several dimension tables. This simple hierarchical relationship eliminates the complexity of nested joins found in traditional normalized schemas, thereby significantly improving query execution speed in analytical reporting systems. Therefore, this study implements the star schema model as the primary analytical data storage structure.

B. Comparative Approach: The Data Vault 2.0 Methodology

Over time, analytical data modeling architectures have continued to adapt to address the challenges of managing large-scale, dynamic data. In addition to traditional models such as the star schema, modern architectural methodologies are being explored to handle the integration of data sourced from multiple operational systems.

Triaji, Subagyo, and Rifai investigated the effectiveness of modeling using the Data Vault 2.0 method in an institutional information system environment [3]. The Data Vault 2.0 architecture has a unique modular structure, in which data is organized into three core components: hubs (storing unique business keys), links (representing relationships between business entities), and satellites (storing historical data along with its descriptive attributes).

Although the Data Vault 2.0 structure tends to be more complex than a typical star schema, this model offers high flexibility and scalability to accommodate changes in operational data schemas without requiring a complete database redesign [3]. Data integration through this modular architecture also supports parallel development processes among development teams, thereby reducing data processing wait times.

C. Data Quality Framework

Before integrated data is loaded into an analytical data warehouse schema, the data quality validation process plays a crucial role in preventing information anomalies. Large-scale data extracted from digital platforms typically carries the risk of quality degradation due to its heterogeneous nature, susceptibility to missing values, and inconsistencies in attribute formats.

Taleb et al. proposed an integrated data quality management framework (Big Data Quality Management Framework) to strengthen data control mechanisms during the pre-processing phase [4]. Within the data governance cycle, data quality assurance must be implemented from the early stages of data cleansing before the data is transformed for final business analysis purposes.

Data quality is assessed based on intrinsic data quality dimensions, including [4]:

- Completeness: Measures the extent to which all relevant data values are recorded without any missing values.
- Consistency: Ensures that values align with the specified format, type, and constraints of the data structure.
- Accuracy: Assesses the accuracy of the data in representing the actual values of the measured objects.

The application of validation rules based on these quality dimensions to the data processing pipeline aims to ensure that the analytical insights generated by the reporting dashboard system have a high level of reliability.

D. Characteristics of the Book-Crossing Dataset

In this study, the experiments are performed on the Book-Crossing data set created by Ziegler et al. [1]. This dataset is composed of the data digital traces of an online reading community, including user profiles, bibliographic metadata of the books, and data instant ratings record. These transactional data are not only the basis for research in the field of recommendation systems, but also reflect the real-world interactions of readers with the books available. Therefore, data cleaning rules in the proposed processing pipeline have to preserve these values, so that they are not deleted in the data transformation phase. This ensures that the results of the descriptive analysis still represent original characteristics of the source community behavior.

III. SYSTEM ARCHITECTURE DESIGN & METHODOLOGY

A. Overall Architecture and ETL Design Rationale

The proposed data pipeline implements the medallion architecture logically, in a single MySQL database instance. The layers are segregated based on the state and purpose of the data which makes smooth connectivity without compromising logical isolation. Reis and Housley describe data flows as one-way flows through the data engineering life cycle. The Raw Layer stores the source data in an immutable CSV format which is the system of record. It has tables like buku, *ratings and users.* The Staging Layer, with tables like *book_stg, rating_stg, and users_stg,* holds cleansed and enriched data, including derived attributes such as Age_group. Finally, the Data Warehouse Layer is built, which is structured in a star schema, with the tables *dim_book, dim_user and fact_ratings*, to answer analytical queries.

An Extract-Transform-Load (ETL) approach was selected over Extract-Load-Transform (ELT) based on three constraints. The source data is of low quality, so loading it into the warehouse as-is would corrupt the aggregate metrics and lead to inaccurate dashboard visualizations. Second, the size of the dataset (111.585 rows) permits an efficient in-memory

processing (null filtering, deduplication, domain validation and surrogate key resolution) with Pentaho Data Integration (PDI). Third, ELT would degrade Power BI dashboard performance, as runtime transformation tasks would compete for computational resources with simultaneous analytical queries. ELT is generally reserved for datasets that are orders of magnitude larger and run on columnar cloud warehouses like BigQuery or Snowflake.

B. Data Pipeline Design

The proposed data pipeline has two main stages: data ingestion and data transformation. In the data ingestion stage, the original Book-Crossing dataset is pulled from the source files and stored in the raw layer without changes. The transformation stage then applies several validation, cleaning, and integration processes before the processed data is loaded into the temporary storage layer. This sequential workflow ensures that only validated data is used to build the data warehouse.

1) Data Ingestion

The raw layer contains the original Book-Crossing dataset that is imported during the data ingestion process. There are three different CSV files in the collection, each one representing a different part of the online reading community: books, users and ratings.

- Book dataset: Provides metadata including International Standard Book Number (ISBN), book titles, authors, publication years and publisher information.
- Users dataset: Contains self-reported age, location strings in localized form and profile information for users across the world.
- Rating data set: This data set contains 111,585 explicit interaction records, which associate a user identifier with a book identifier and the rating value of that interaction.

Each dataset has a separate Pentaho Data Integration (PDI) transformation. The workflow starts with a CSV File Input step to read records from the source file and concludes with a Table Output step to load the extracted data into the corresponding raw table in MySQL. Isolating the extraction process for each dataset simplifies future system maintenance and allows each data source to be processed independently according to its unique structure.

To enable this isolation, we configure the Table Output step with the Truncate Table option, creating a comprehensive full-load strategy that captures the latest state of the source dataset. In addition, the pipeline has a built-in feature that allows it to automatically delete rows in the raw table before inserting new records to avoid stale data or duplicate records being left behind in the database. This is consistent with the nature of the Book-Crossing dataset, which is distributed as a complete snapshot, rather than incremental updates.

The current implementation uses this full-load method, but the pipeline architecture is designed to support future extensions for incremental loading. Such future improvement can be achieved by adding an upsert feature using Merge Row transformations and watermark columns, without any changes to the downstream transformation processes or the basic design of the data warehouse. After this ingestion phase successfully creates a clean and complete snapshot within the raw layer, the data is immediately pushed into the next phase of the pipeline, the data transformation stage.

2) Data Transformation

When the source data is successfully loaded into the raw layer, the book, user and rating datasets are each processed by three independent transformation workflows before loading into the preparation layer. The aim of this stage is to improve the quality of the data by applying rules of cleaning, validation and integration, so that only validated records proceed to the next stage of the workflow.

The book transformation is meant to improve the consistency of the book metadata. We drop records with missing ISBN values as the ISBN is used as the primary identifier throughout the pipeline. Duplicate ISBN records are also removed so that one book is represented only once before it is stored in the staging layer.

The Age attribute is used by User transformation to verify demographic data. Any age values outside the range of 5 to 100 years are considered invalid. Replaced with NULL. The users are then validated. They are classified in the categories of Teen, Young Adult, Adult, Senior and Unknown. This is done through a field called `Age_Group'. The age groups are assigned based on the age values. For example users with age 5-19 are considered Teen. Users with age 20-39 are considered Young Adult. Users with age 40-64 are considered Adult. Users with age 65-100 are considered Senior. If the age is not, in the range of 5-100 years then it is classified as Unknown. This extra attribute makes it easier to analyze the demographic data when displayed on the dashboard.

In the rating transformation, duplicate User ID and ISBN combinations are removed to keep a distinct record of each user-book interaction. However, as this indicates user interaction and not erroneous data, a rating value of 0 is maintained. The original interaction characteristics of the Book-Crossing community are preserved in the processed dataset by retaining these records.

The modified book, user, and rating datasets, after all the validation and cleaning procedures are completed, are combined with their respective identifiers and imported to the staging layer. This integration guarantees referential consistency between related entities and generates a consistent intermediate dataset, which is the foundation for building dimensional models in the data warehouse.

To make sure the changed data is good for analysis, the plan uses a structured framework to check the data quality while it is being changed. This framework does not just use one rule to check the data. It checks the data in a sequence to make sure it is consistent follows the rules and is accurate. The framework checks the data in layers to improve the reliability of the data before it is put into the data warehouse. You can see a summary of the framework in Table I. The data quality validation

framework is important for the transformed data. The transformed data needs to be checked to make sure it is good, for analysis.

TABLE I. THREE-LEVEL DATA QUALITY VALIDATION FRAMEWORK

Validation Level	Validation Type	Validation Rules Applied
Level 1	Structural Validation	Records with missing ISBN values are removed, and duplicate records are eliminated using the *Unique Rows (HashSet)* step in *Pentaho Data Integration (PDI)*.
Level 2	Domain Validation	Book-Rating values are preserved within the range of 0–10, while Age values outside the range of 5–100 are replaced with NULL. A new Age_Group attribute is then derived, classifying users into *Teen*, *Young Adult*, *Adult*, *Senior*, or *Unknown*.
Level 3	Referential Integrity Validation	Foreign key relationships between *Fact_Ratings*, *DimBooks*, and *DimUsers* are validated using Database Lookup. Records without matching dimension entries are excluded before loading into the data warehouse to maintain referential integrity.

C. Data Warehouse Design

Figure 1 illustrates the dimensional model applied in the proposed data warehouse. After the data sets are validated and integrated in the staging layer, the data is loaded into a centralized MySQL data warehouse organized using a star schema. This architecture was used in order to reduce data redundancy and improve the efficiency of analytical queries by separating transactional data from descriptive information. Additionally, the multidimensional analysis needed for dashboard visualization and business intelligence is supported by the streamlined interaction between the fact table and the dimension tables.

The star structure implemented has one fact table and two dimension tables that are linked by a one-to-many relationship.

- Fact_Ratings stores transactional data, including user ratings and references to the relevant book and user records.
- DimBooks contains descriptive information about each book, including the ISBN, title, author, publication year, and publisher.
- DimUsers stores user demographic information, including location, verified age, and the derived Age_Group attribute.

This procedure divides transactional data into fact tables and descriptive qualities into dimension tables. This allows the successful integration of data from several dimensions during reporting, and allows for analytical queries. The dimension tables employ auto-incrementing integer surrogate keys (book_sk and user_sk) as primary keys in order to accomplish this structural separation, maximize the data warehouse's performance, and preserve its independence from operational data sources. The natural keys of the original User-ID and ISBN are preserved for reference and search. Such architecture enables MySQL to join tables with integer-based foreign keys more effectively than alphanumeric strings and contains the downstream impact of changes in source identifiers.

The load_fact_ratings transformation solves these relationships in the last loading phase simultaneously. It makes two sequential lookups to the database to get the surrogate key of each incoming ISBN and User-ID and inserts records in Fact_Ratings. In the final insert process, transactional records without a dimension entry are strictly excluded to ensure total referential integrity in the data warehouse.

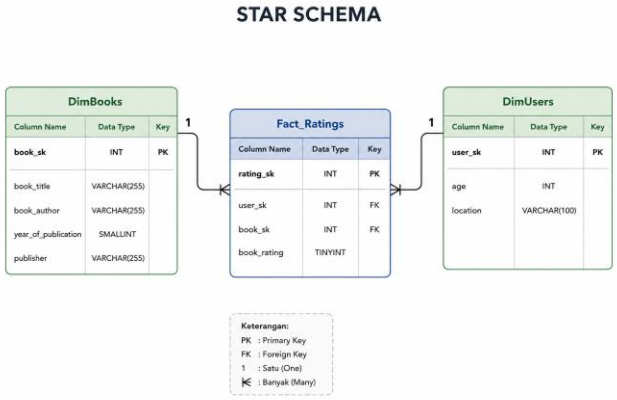


Fig. 1. Star Schema

D. Workflow Automation

To make the ETL process work better and ensure that everything runs smoothly the proposed pipeline uses a Pentaho Data Integration job, which is called main_job_library.kjb. This job makes sure that all the transformations happen in the order so the whole pipeline can run automatically. By putting all the workflow into one job it becomes easier to manage. There is less chance of making mistakes when running it manually.

The workflow starts by clearing out all the tables in the staging and data warehouse layers. This step makes sure that each time the pipeline runs it starts with a slate before new data is added. After the database is ready three data extraction transformations run at the time to import the book, user and rating datasets into the raw layer. Since these datasets do not depend on each other running them at the time makes the process faster without affecting the quality of the data.

Once the raw layer has all the data three transformation workflows run to clean up check and add value to the data before loading it into the staging layer. After all the staging processes are done the data pipeline moves on to load the dimension models. The load_dim_books and load_dim_users transformations run first because they create the keys that the fact table needs. Only after both dimension tables have data does the pipeline run load_fact_ratings, which does the database lookups before adding records to the fact table. The workflow ends with a "Success" note, which means that all the ETL processes have finished successfully.

This workflow uses a strategy that always produces the result no matter how many times it is run, by clearing out the tables and loading all the data. This makes the pipeline more reliable and easier to maintain especially when the ETL process needs to be run after something goes wrong or when new data becomes available from the Book-Crossing dataset. The ETL process and the pipeline work together to make sure that the data warehouse is always in the state, which is important, for the ETL process and the pipeline.

IV. IMPLEMENTATION

A. Pipeline Tool Configuration

The proposed ETL pipeline is implemented using Pentaho Data Integration (PDI) Community Edition and MySQL as the database management system. This implementation is based on the three-tier architecture as described in Section III which includes the raw data layer, the staging layer and the data warehouse. The whole ETL process is coordinated by one Pentaho Job, so the whole workflow can be run automatically from data extraction to loading into the data warehouse.

This implementation contains nine transformations, like data ingestion, data transformation, dimension loading, and fact loading. The whole system is a single automated workflow but each transformation is a separate module which makes it easier to maintain and modify in the future. The modularity ensures that the data pipeline will function consistently and the processed data will be available for consumption in the data warehouse.

B. Warehouse Loading and Surrogate Key Resolution

The MySQL data warehouse successfully loads the validated data from the staging layer. The data warehouse is designed as a star schema with two different dimension tables (DimBooks and DimUsers) and a single fact table (Fact_Ratings) in the middle. This is a dimensional modeling technique that separates transactional metrics from descriptive master data, providing the basis for extremely effective analytical queries.

The warehouse keys for this dimensional model can only be built in this order so the pipeline must first populate the dimension tables before inserting the fact table due to structural dependencies. The fact table loading occurs following the successful loading of dimension tables and the production of the auto-incrementing integer surrogate keys (book_sk and user_sk). During this phase the original alphanumeric ISBN and User-ID values stored as natural keys are converted to the appropriate surrogate keys using the Database Lookup procedures. During this last insertion process, transactional records that do not have a matching dimension of entry are strictly filtered out and removed from Fact_Ratings. The mechanism guarantees strict referential integrity in the analytical repository, reduces the impact of structural changes in the source system, and optimizes MySQL join performance using integer-based foreign keys.

C. Dashboard Development

The single source of truth for the interactive analytical dashboard built in Power BI is the fully populated star schema in the data warehouse. The visualization platform connects directly to the MySQL database through an Open Database Connectivity (ODBC) driver to import the pre-validated dimensional data for business intelligence processing. This is ensured by strictly limiting the data retrieval to the data warehouse layer rather than querying the raw or the staging layers. All the generated visual elements are thus entirely decoupled from the data quality anomalies and processing overhead of the upstream pipeline phases.

This clean data foundation drives the dashboard, which offers a rich set of interactive visualizations for multi-dimensional analysis of the Book-Crossing community. The visualizations are: geographical distribution of the readers, top rated books, author ranking measures based on explicit rating averages, overall score distributions, and demographic segmentations based on the calculated Age_Group attribute. Also, the use of interactive runtime filters for publication years, specific publishers and age groups allows users to explore data trends from different analytical perspectives. Such an architecture allows for flexible data exploration on the reporting interface without the need to perform resource-intensive transformations, nor to make structural changes in the underlying data warehouse schema.

V. RESULT AND DISCUSSION

A. Dashboard Overview

This Power BI dashboard shows some Book-Crossing data with interactive visualizations and four KPI cards. The dashboard provides a snapshot of key data on book popularity, author performance, rating trends, reader demographics and geographic distribution. By using interactive filters for Age Group, Publisher, Year of Publication, users can view this data from many different perspectives.



Fig. 2. Power BI dashboard: Book-Crossing Community Insights.

B. *Analytical Findings*

Analytical Results The processed dataset consists of 111,585 ratings from 28,909 users for 496 books with an average rating of 3.28. Although this average rating appears low, the distribution of ratings suggests that this is strongly influenced by the large number of entries with a rating of 0, which in the Book-Crossing dataset represent implicit user interactions, not negative ratings. Therefore, the average rating should be seen as a property of the dataset, not as a quality measure of the books.

This dashboard also gives different views on reader engagement. The most reviewed book was Wild Animus. The Lovely Bones and The Da Vinci Code came in second and third, respectively, showing that these are the most popular titles among the community. But popular doesn't always translate into more appreciations from the reader. This is shown in the "Most Popular Authors" and "Authors with the Highest Average Ratings" visualizations. More reviews for authors like John Grisham and James Patterson and better average ratings on fewer reviews for Sandra Brown and Patricia Gaffney. The results are consistent with the number of reviews being a proxy for reader engagement . while average ratings are better indicators of how much readers appreciate an author's work.

Demographically, the Adult age group (31-60 years) made up 47.84% of the users analysed, followed by the Young Adult, Teen and Senior groups. The geographic visualisation also shows that the readers come from many different countries, with a higher density in North America and Western Europe. Overall, these results show that the developed data warehouse supports the multidimensional analysis of the Book-Crossing dataset by unifying information about books, authors, demographics and geography in a single analytics platform.

C. *Discussion: Strengths and Limitations*

The established ETL pipeline demonstrates that a well organized data engineering approach enhances a dataset's usuability for subsequent analysis. The logical division of the pipeline into layers aids in preserving he workflow and in tracking the data distinctly. This architecture enables the auditing of the transformation lineage from the original raw files to the final analytical tables when updates or reviews are necessary.The pipeline shifts data quality management to the ETL phase, ensuring that only verified data is stored in the warehouse instead of addressing data quality problems at the visualization phase. Automated cleaning rules are implemented prior to insertion to eliminate duplicate entries, rectify invalid values, and uphold consistency standards. As a result, the Power BI dashboard functions solely on pre-processed data, minimizing runtime calculation load and delivering reliable analytical outcomes.

The star schema used is effective in supporting multidimensional analysis by separating transactional metrics from descriptive attributes. This structure allows easy filtering and aggregation of data, and offers a responsive foundation for dashboard exploration. Although this study did not quantitatively measure query execution times, the data warehouse successfully supported all visual configurations without encountering functional bottlenecks.

Nevertheless, certain system limitations must be noted. The pipeline is a full-load approach, so the dataset needs to be reloaded completely for each run. While sufficient for the current volume, this approach is inefficient for larger or frequently updated datasets. Moreover, this analysis was conducted on a subset of 496 books and not on the whole Book-Crossing dataset, hence the results are limited to this specific data.

In the future, we will work on creating the incremental loading mechanism so that it pulls only new or changed records using watermark columns. Expanding the warehouse to encompass the complete dataset will also broaden the analytical scope. Finally, formal performance testing for assessing query responsiveness with larger data volumes is needed to assess the overall scalability of the proposed workflow

VI. CONCLUSION

This paper presented the design and implementation of an automated, repeatable ETL pipeline for the Book-Crossing dataset utilizing Pentaho Data Integration, MySQL, and Power BI. The architecture has been designed in three distinct logical layers (raw, staging and data warehouse). The system has been able to automate the extraction, transformation and loading processes. Our pipeline proved to be operationally reliable , as it successfully processed the source data into a validated dimensional model comprising 111,585 interaction records, 496 unique books and 28,909 verified users. This pre-selected repository is the primary data source for an interactive Power BI dashboard. This allows for multi-dimensional analysis, including detailed insights into reader geographic distribution, author performance, and book popularity.

The current design meets all functional requirements of the integrated subset. There is a possibility to improve the system further by expanding the use case. To maximize resource consumption, we will switch from a complete load strategy to an incremental loading technique in the following steps. We also recommend expanding the data warehouse capacity to include the entire, unrestricted dataset, and conducting formal

query latency stress tests to evaluate the scalability and performance of the proposed data engineering approach.

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REFERENCES

[1] C.-N. Ziegler, S. M. McNee, J. A. Konstan, and G. Lausen, “Improving recommendation lists through topic diversification,” in Proc. 14th Int. Conf. World Wide Web (WWW ’05), Chiba, Japan, 2005, pp. 22–32.

[2] R. Kimball and M. Ross, The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling, 3rd ed. Indianapolis, IN: Wiley, 2013.

[3] S. Feng and L. Zheng, "Research on Data Warehouse Modeling Based on Star Schema," in Proc. 3rd International Conference on Electronic Information Technology and Computer Engineering (EITCE), Xiamen, China, 2019, pp. 1152-1155, doi: 10.1109/EITCE47263.2019.9870050.

[4] I. Taleb, R. Dssouli, and M. A. Serhani, "Big data pre-processing: A quality framework," in Proc. IEEE International Congress on Big Data, New York, NY, USA, 2015, pp. 191-198, doi: 10.1109/BigDataCongress.2015.34.